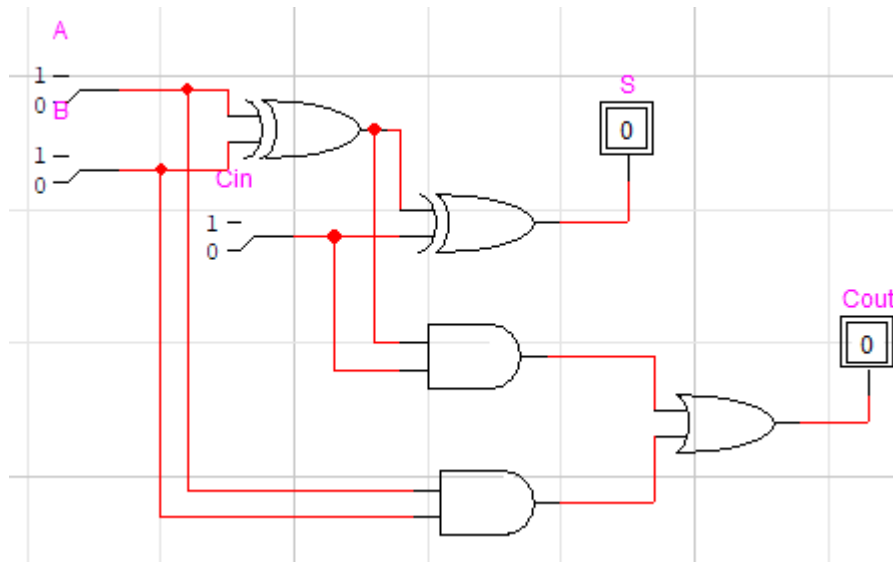


Task 1:

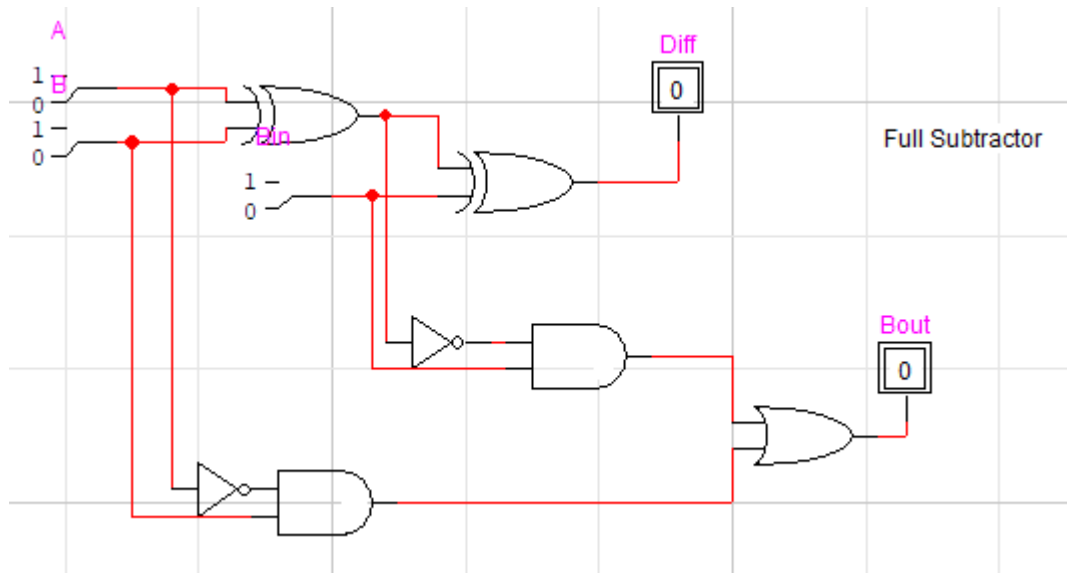
Full Adder:-

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

**Task 2:**

Full Subtractor:-

A	B	Bin	D	Bout	A'
0	0	0	0	0	1
0	0	1	1	1	1
0	1	0	1	1	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	0	0	0
1	1	1	1	1	0

**Task 3:**

For Sum:-

AB \ Cin	0	1
00		1
01	1	
11		1
10	1	

$$A'B'Cin + A'BCin' + ABCin + AB'Cin'$$

$$S = A \oplus B \oplus Cin$$

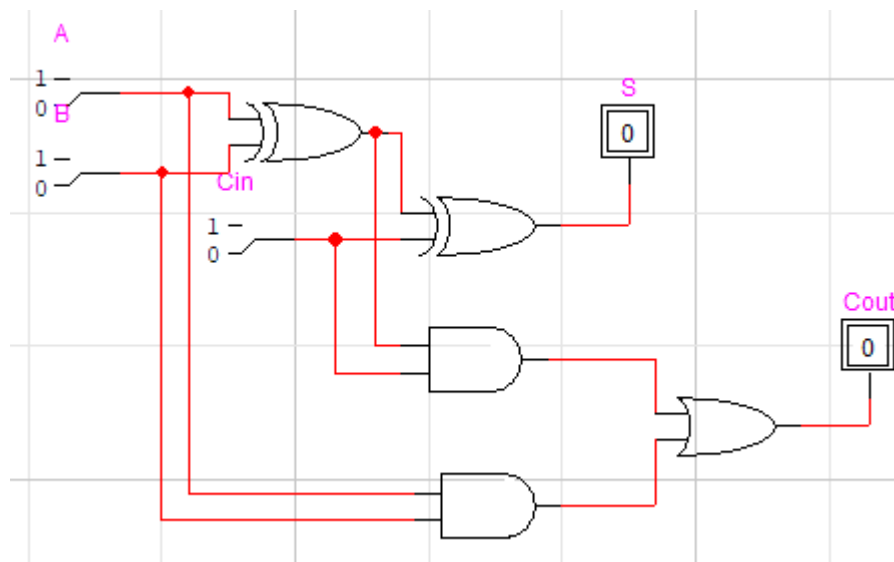
A	B	Cin	S
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

For Carry:-

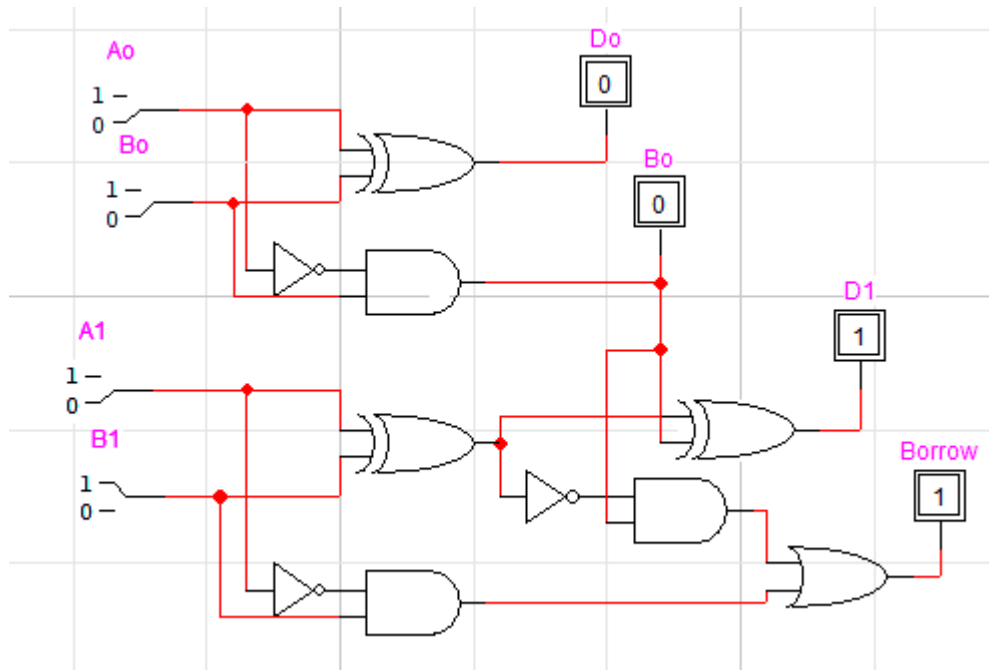
AB \ Cin	0	1
00		
01		1
11	1	1
10		1

$$BC_{in} + AB + AC_{in}$$

A	B	Cin	Cout
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1



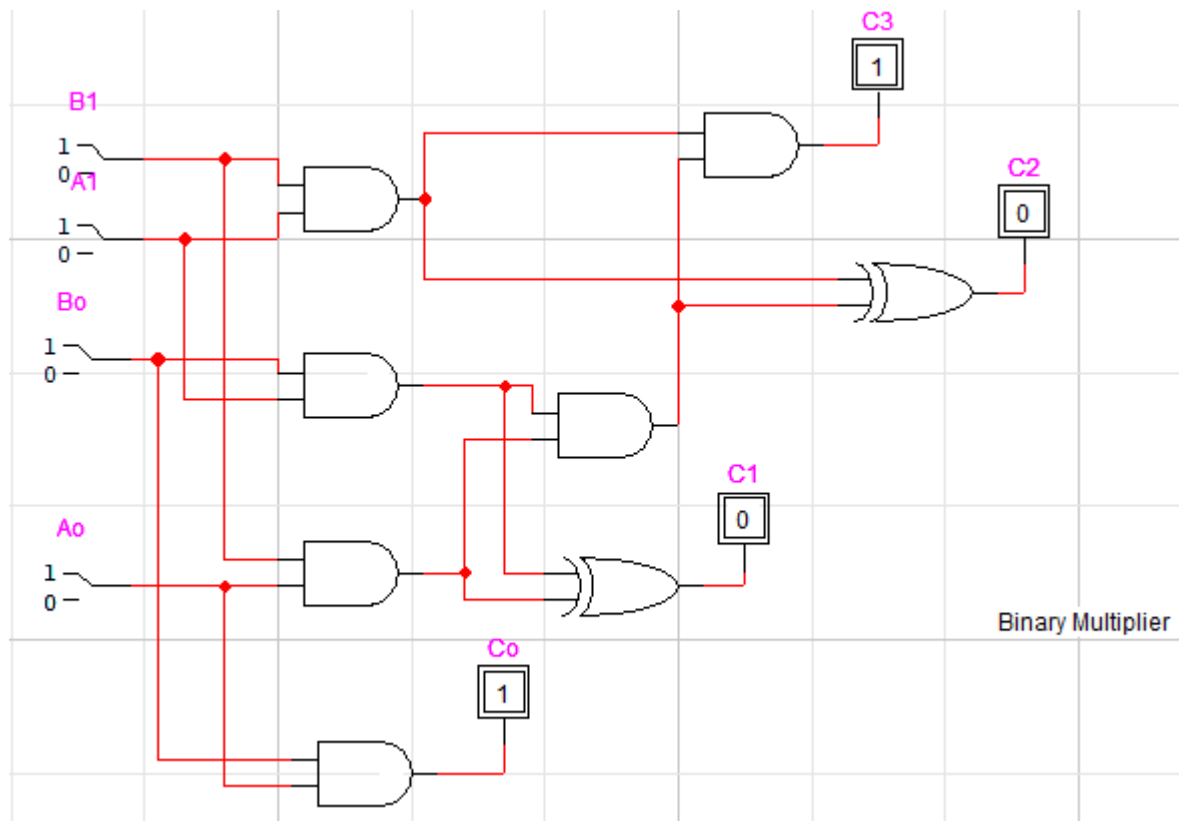
Task 4:



A ₁	A ₀	B ₁	B ₀	D ₁	D ₀	Borrow
0	0	0	0	0	0	0
0	0	0	1	1	1	1
0	0	1	0	1	0	1
0	0	1	1	0	1	1
0	1	0	0	0	1	0
0	1	0	1	0	0	0
0	1	1	0	1	1	1
0	1	1	1	1	0	1
1	0	0	0	1	0	0
1	0	0	1	0	1	0
1	0	1	0	0	0	0
1	0	1	1	1	1	1
1	1	0	0	1	1	0
1	1	0	1	1	0	0

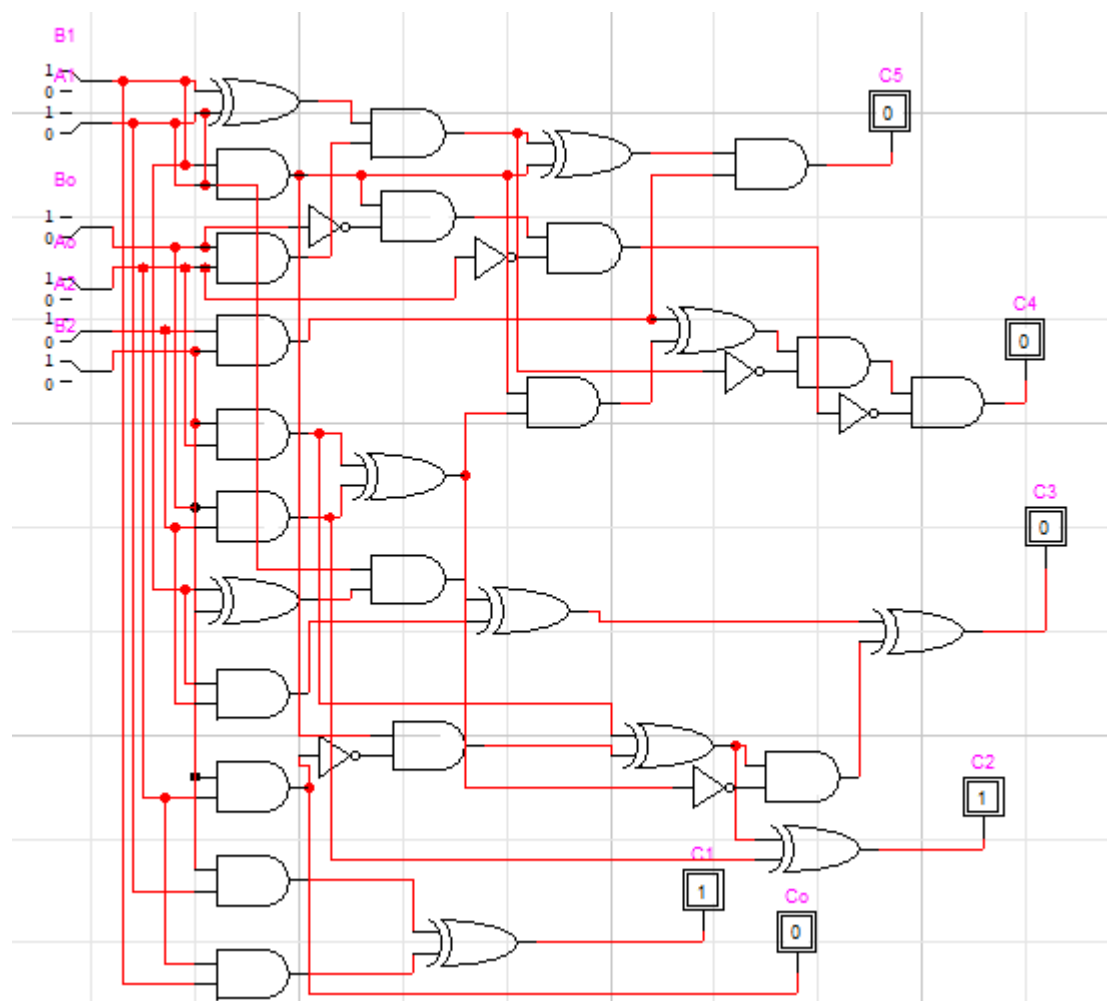
1	1	1	0	0	1	0
1	1	1	1	0	0	0

Task 5:



A ₀	A ₁	B ₀	B ₁	C ₀	C ₁	C ₂	C ₃
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	0
0	1	0	1	0	0	0	1
0	1	1	0	0	0	1	0
0	1	1	1	0	0	1	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	1	0
1	0	1	0	0	1	0	0
1	0	1	1	0	1	1	0
1	1	0	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	1	1	0
1	1	1	1	1	0	0	1

Task 6:

**Task 7:**

